HACKATHON | PROBLEM STATEMENT 18

Team: cyber knight

Fish Disease Risk Detection & Early Warning!

Smart AI early warning system that catches fish illness early and protects pond health

THE PROBLEM

- **Late Detection:** By the time sores, red spots, or pale gills appear on fish, the illness has already spread to the whole pond.
- **Disconnected Data:** Fish photos, water tests (oxygen, pH, ammonia), and farm loss logs sit in separate places with no connection.
- **Heavy Fish Losses:** Acting too

OUR SOLUTION

📷 Pond Fish Photos
Spots 7 fish diseases (sores, fins, gills, scales)

🐟 Fish Behaviour Logs
Tracks surfacing, lethargy & slow feeding activity

📋 Mortality Records
Saves daily losses & past pond checkup history

🧪 Pond Water Tests
Measures Oxygen, pH, Temp & toxic Ammonia

▼

AI Risk Engine: CNN Photo AI + Water Model + Disease Synergy

WHAT WE WILL BUILD

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Features, Technology & Impact

Simple, practical tools designed for real fish farmers to protect crops without antibiotics

CORE FEATURES

- ✓ **Phone Photo Scanner:** Take a quick photo of sick fish to detect 7 major diseases (Red sores, Dropsy, Gill disease, White tail, Fungus, Parasites) with 98% accuracy.
- ✓ **Water Quality Checker:** Checks Oxygen, pH, Temperature, and Ammonia levels in seconds to flag dangerous pond water

ADVANCED FEATURES

- ★ **Natural Dosage Calculator:** Enter pond length, width & depth to calculate the exact kilograms and 50 kg sacks of salt and lime needed—no guessing.
- ★ **Past Pond Health Records:** Keeps a clean digital ledger of every test, photo, and treatment so farmers can track trends across all ponds.

TECH STACK & IMPACT

SIMPLE TECH STACK

AI / ML: Python · MobileNetV3 CNN · Random Forest · OpenCV

Backend: FastAPI · Python 3.12 · Uvicorn · Supabase Cloud

Frontend: React 19 · Vite · Tailwind CSS · Fast Mobile Web